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ORIGINAL PAPER

High levels of human chitotriosidase hinder the formation of peritrophic membrane in anopheline vectors

M. Di Luca · R. Romi · F. Severini · L. Toma · M. Musumeci · A. M. Fausto · M. Mazzini · G. Gambellini · S. Musumeci

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Abstract In the *Anopheles* midgut, *Plasmodium falciparum* produces a specific chitinase able to penetrate the blood meal surrounding the chitin-containing peritrophic membrane (PM). High levels of an analogous chitinase, chitotriosidase (CHIT), may be found in human blood, being the markers of macrophage activation. To verify the hypothesis that CHIT present in malaria patient blood could help parasite to overcome PM, we carried out a bioassay by feeding *Anopheles stephensi* females on an artificial apparatus that contained human blood from four different sources and with different chitinase concentrations: (1) healthy donors, as negative controls; (2) patients with malaria; (3) patients with Gaucher disease; and (4) whole blood enriched with commercial *P. falciparum* chitinase, as positive controls. After 16, 20 and 24 h of bloodfeeding, mosquitoes were dissected to extract the midgut and assess the effect of the different chitinases on membrane structure. Optical microscopy showed that formation of PM was clearly complete after 16 h in the posterior midgut from *Anopheles* already fed with healthy donor bloods. By contrast, PM formation was visible after 16 h in the posterior midgut of mosquitoes fed with malaria and Gaucher patient bloods but appeared clearly damaged at 20 and 24 h. At the same time, the PM formation was almost completely inhibited in the midgut of *Anopheles* fed with *P. falciparum* chitinase-enriched bloods. These alterations were clearly confirmed by transmission electronic microscopy. In the present paper, we demonstrate that human CHIT from different sources is active on anophelines' PM.

M. Di Luca · R. Romi · F. Severini · L. Toma · M. Musumeci
Department of Infectious, Parasitic and Immune-Mediated Diseases, Istituto Superiore di Sanità,
Rome, Italy

A. M. Fausto · M. Mazzini
Department of Environmental Sciences, University of Tuscia,
Viterbo, Italy

G. Gambellini
Interdepartmental Centre of Electron Microscopy,
University of Tuscia,
Viterbo, Italy

S. Musumeci (✉)
Department of Pharmacology, Obstetrics and Gynecology,
Pediatrics, University of Sassari,
Viale San Pietro 43b,
07100 Sassari, Italy
e-mail: smusumeci@tiscalinet.it

S. Musumeci
Institute of Population Genetics,
Italian National Research Council,
Alghero, Italy

Introduction

Chitinases, largely diffused in nature, are found in prokaryotes, eukaryotes (Flach et al. 1992) and in several pathogens. Their biological roles include the hydrolytic attack to host cell walls. Analogous chitinase is produced by *Plasmodium falciparum* to invade the *Anopheles* midgut by crossing the peritrophic membrane (PM). Therefore, *P. falciparum* chitinase could be a potential target for blocking transmission of parasite from human host to the mosquito vector (Huber et al. 1991; Li et al. 2005). Recently, it has been discovered that humans produce an enzyme showing chitinase activity: the chitotriosidase (CHIT).

Its concentration has been found up to two orders of magnitude larger in plasma of patients suffering by Gaucher disease, a rare genetic disorder caused by a mutation in the glucocerebrosidase gene (Giraldo et al. 2001). This enzyme

is active against colloidal chitin and is inhibited by the chitinase inhibitor allosamidin.

Elevated CHIT activity has surprisingly been found also in patients with acute and severe *P. falciparum* malaria (Barone et al. 2003). Moreover, a positive correlation was also found between plasma CHIT and serum ferritin levels (Musumeci et al. 2005; Bouzas et al. 2003) in such patients. Bouzas et al. (2003) reported that an accelerated red cell turnover, associated to macrophage accumulation of red cell membrane degradation products, may induce macrophages to produce CHIT (Bouzas et al. 2003). Anopheline mosquitoes introduce in the midgut elevated quantities of CHIT, produced by *P. falciparum*-activated macrophages, when ingesting a blood meal from patients with acute malaria. In the present paper, we demonstrate that human CHIT from different sources is active on anophelines' PM, probably enhancing the penetration action of malaria parasites in the mosquito midgut.

Materials and methods

Mosquito strain A long-established laboratory mosquito strain of *Anopheles stephensi*, maintained at the Istituto Superiore di Sanità (ISS) as a reference strain for susceptibility/resistance to insecticides, was reared in a climatic chamber at 26±1°C, 70–80% relative humidity and a photoperiod of 16:8 h light/dark.

Adults were maintained in 35×35×35 cm steel frame cages, covered by a cotton net of 96 meshes/cm^2 and provided ad libitum with a solution of 5% glucose both before and after bloodfeeding. Twenty-four hours before bloodfeeding, the glucose solution was replaced with distilled water to encourage engorgement.

Membrane feeding assay Seven batches of 30 5- to 7-day-old mosquito females were placed in 500-ml proof cartons, closed with a 96 mesh/cm^2 cotton net, without food and fed through a membrane feeding apparatus. Blood samples were kept warm to 37°C and transferred to five chambers of a Hemotek feeding apparatus (Discovery Workshops, Accrington, Lancashire, UK) in which the original Hemotek membrane was replaced with a 5×5 cm piece of fresh pork gut. Chambers were placed on top of the proof cartons containing the mosquito batches. After a half-hour feeding period, fully engorged females were transferred to small cages in the climatic chamber and maintained with 5% glucose solution until dissection.

Blood samples Seven blood samples were considered for the bioassay: (a) two negative controls, represented by a healthy donor human 0 Rh-negative bloods (samples 1 and 2) showing a CHIT activity of <0.10 mU/ml; (b) three positive controls (samples 3–5), represented by human blood added with exogenous *Plasmodium falciparum* chitinase (PfCHT1; New England Biolabs) to a final concentration of 100 mU/ml, 500 mU/ml and 2000 mU/ml, respectively; (c) two positive controls (samples 6 and 7, with 5.2 and 10.4 mU/ml CHIT activity, respectively), from patients affected by Gaucher disease, the most common lysosomal storage disease that is characterized by production of high chitotriosidase levels; (d) two acute *P. falciparum* malaria African patient samples (samples 8 and 9) having a blood chitotriosidase activity of 0.68 and 0.95 mU/ml, respectively. The 0 Rh-negative human blood was used to lead samples 3–5 to a final volume of 2 ml and a hematocrit of 50:50 serum/cells.

Briefly, CHIT activity was measured by incubating 5 μl of sample with 0.1 ml of a solution containing 22 μmol/l of the fluorogenic substrate 4-methylumbelliferyl-β-D-*N*,*N*′,*N*″-triacetyl-chitotriose (Sigma Chemical n. catalog. M5939) in 0.5-M citrate-phosphate buffer pH 5.2 for 15 min at 37°C, as originally described by Hollak et al. 1994. The reaction was stopped by addition of 2 ml of 0.5 mol/l Na_2CO_3–$NaHCO_3$ buffer, pH 10.7. The fluorescence of 4-methylumbelliferone was read on a spectrofluorimeter Hitachi 2500 (Hitachi, Europe, Herts, UK), on 365-nm excitation and 450-nm emissions. Chitinase activity was expressed as nanomoles of substrate hydrolyzed per milliliter per hour (nmol^{-1} ml^{-1} h) or expressed in mU/ml considering one unit of 4-methylumbelliferyl-β-D-*N*,*N*′,*N*″ triacetyl-chitotriose-degrading activity as one unit of fluorescence emission under the assay conditions.

Preparation and examination of histological sections Blood-fed mosquitoes were anaesthetized with CO_2 and dissected (ten specimens/batch) to extract the midgut at three different post-bloodfeeding times (16, 20 and 24 h). Ten guts were taken at each time point for each treatment (i.e. a total of 210 midguts). Intact midguts, including entire blood-meal contents, were dissected into phosphate saline buffer (pH=7.2) and immediately fixed for 1 h at 4°C with 4% paraformaldehyde and 5% glutaraldehyde in 0.1-M cacodylate buffer (pH 7.2). After stirring overnight in the same buffer, samples were post-fixed in a 1% osmium tetroxide in 0.1-M cacodylate buffer for 1 h. Specimens were dehydrated in a graded ethanol series and embedded in Spurr resin. Thin sections were cut with Reichert Ultracut and LKB Nova ultramicrotomes using a diamond knife. Semithin sections (1 μ thick) collected on slides were stained with toluidine blue and then observed with Zeiss Axiophot optical microscopy (OM). For transmission electronic microscopy (TEM) measurements, ultra-thin sections (60–70 nm thick) were collected on copper grids, stained with uranyl acetate and lead citrate and observed with a JEOL 1200 EX II electron microscope. To obtain

reproducible histological sections, cuts were made towards the posterior of the posterior midgut because it is in this region that PM forms most prominently.

Both OM and TEM images were digitalized, and the PM thickness was accurately measured using the Image Tool program (Ver 3 for Microsoft Windows NT).

Results

OM morphological examination of untreated samples 1 and 2, 16 h after bloodfeeding, revealed a detectable PM (Fig. 1a). PM thickness increased at 20 and decreased at 24 h (Fig. 1b,c). PM has been recognized as a fibrous and laminated structure that encloses and surrounds the entire blood meal, just above the epithelium. Heme precipitates and matrix-bound heme are clearly visible between the erythrocytes at 20 h. TEM observation confirmed a perfectly maintained PM structure, showing an electron-dense layer that separates the blood meal (bolus of erythrocytes) from the microvilli (Fig. 2a). During the blood digestion, the PM internal portion became filled by heme precipitates (Fig. 2b,c) although maintained both compact structure and barrier function.

Both OM (not shown) and TEM data, concerning midgut from *Anopheles* fed with blood from malaria patients, indicated delayed and damaged PM formation, according to CHIT concentration (0.68 and 0.95 mU/ml of CHIT activity, respectively).

Figure 2d–f shows the midgut section from *Anopheles* fed with blood from patient with the higher CHIT activity (0.95 mU/ml). PM appears only partly conserved at 20 h (Fig. 2e) and completely filled by heme precipitates, which in some areas directly contact microvilli of epithelial cells, at 24 h (Fig. 2f).

Also in *Anopheles* fed with blood of patients with Gaucher disease (5.2 and 10.4 mU/ml of CHIT activity, respectively), the PM thickness was markedly reduced. In Fig. 3, the midgut section from *Anopheles* fed with blood from patient with the lower CHIT activity (5.2 mU/ml) is shown. PM appears filled by products of digestion since 16 h after blood meal (Fig. 3a). A PM thin fibrillar portion divided the blood products from microvilli during the first two digestion phases (Fig. 3a,b). At 24 h, in some areas, heme precipitates are found in contact with the microvilli layer (Fig. 3c).

In the commercial chitinase, from *P. falciparum* (100, 500 and 2,000 mU/ml of PfCHT1 activity, respectively), enriched blood samples, the PM appears clearly disrupted, and the heme precipitates reached the epithelium. These alterations were clearly observed by TEM at 16, 20 and 24 h. Figure 3d–f shows midgut sections of the sample obtained by adding 2,000 mU/ml of PfCHT1. Since 16 h, PM appears as a thin fibrillar structure acting as a barrier only in a few areas (Fig. 3d). Subsequently, at 20 and 24 h after blood meal, PM seems disorganized and completely permeable to blood digestion products, clearly visible among the microvilli and inside the epithelial cells (Fig. 3e–f).

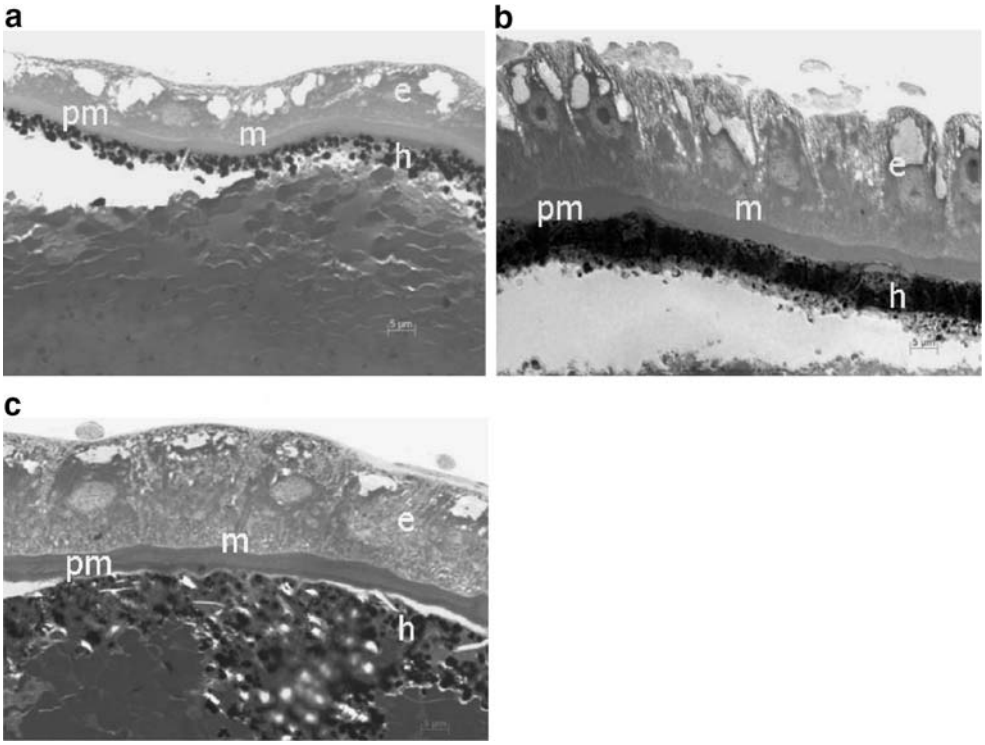

Fig. 1 Morphological examination of the midgut from *Anopheles* fed with blood of healthy donor reveals a fully developed PM by 16 h (**a**), whose thickness increases at 20 h and decreases at 24 h (**b** and **c**)

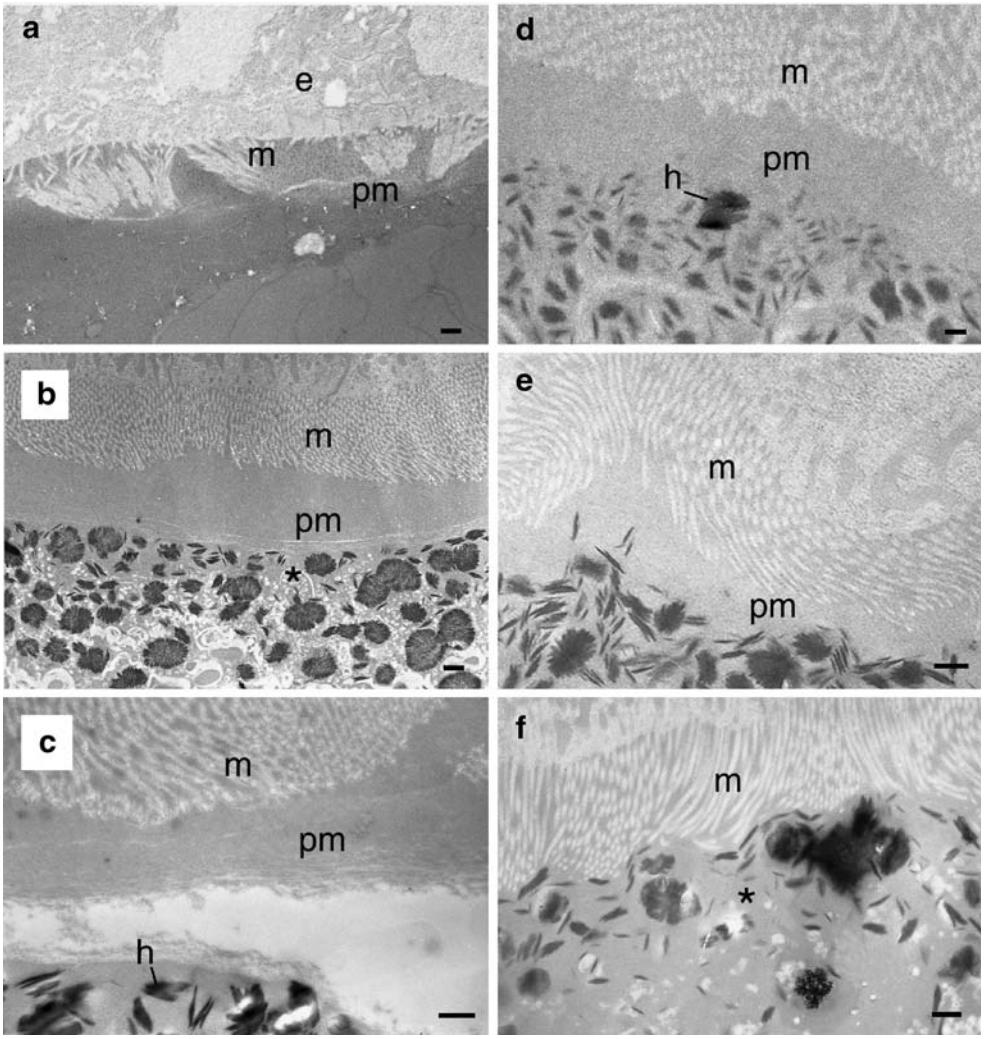

Fig. 2 TEM micrographs of midgut sections from *A. stephensi* female mosquitoes at 16 h (**a** and **d**), 20 h (**b** and **e**) and 24 h (**c** and **f**) after blood-feeding. Scale bar=500 nm. **a**–**c** Mosquitoes fed with blood of healthy donors. PM is well formed, showing an electron-dense laminated structure interposed between the microvilli (*m*) of midgut epithelial cells (*e*) and the blood meal (*asterisk*). During the blood digestion, the PM appears structurally conserved and continues to maintain a barrier function. *h* Heme precipitates. **d**–**f** Mosquitoes fed with blood containing 0.95 mU/ml CHIT from malaria patient. PM is partly conserved and appears completely filled by heme precipitates (*h*) that, in some areas, directly contact microvilli (*m*) of epithelial cells. Blood meal (*asterisk*)

In Fig. 4, the progression or regression of PM in presence of Chit, as measured on the microscopic section of stomach (PM/epithelium), is graphically reported. In fact, in the midgut of *Anopheles* fed with blood of malaria patients and of Gaucher patients, the PM formation was measurable after 16 h but clearly reduced at 20 and 24 h in a degree proportional to Chit contents. These alterations were clearly documented at TEM at 16, 20 and 24 h (see Fig. 2). Now, most of heme bound to the PM is associated with enormous number of small electron-dense granules, and the residual PM is full of erythrocytes and heme precipitates, also if the PM could appear structurally conserved. This alterations were visible in samples of mosquitoes that were fed with blood containing 5.2 mU/ml of chitinase from malaria patient and 2,000 mU/ml of chitinase from *P. falciparum* at different magnitude (Fig. 3). In the sample added with chitinase from *P falciparum*, the alteration of PM were clearly visible, and the products of blood digestion including heme precipitates (h) are clearly visible among the microvilli (m) and inside the epithelial cells (e) since the 20 h (see Fig. 3). At the endoperitrophic side, some of these granules seem to dissociate from the PM and leak to the lumen.

Discussion

Anopheline mosquitoes, as most of insects, produce a PM permeable to gastric juice that separates the blood meal from stomach epithelium. Diffusion of hydrolytic enzymes from the midgut epithelium across the PM is essential for blood digestion. In principle, a PM absence could permit to digestive enzymes a quick access to the food bolus, thus resulting in faster digestion and, consequently, better absorption of nutrients. Nevertheless, the PM's main task is the protection of the epithelium cells from hard/wrinkled undigested particles or foreign organisms. Thus, in the evolution, the slight disadvantage caused by a small decrease in the digestion rate has been compensated by the higher protection afforded by PM (Villalon et al. 2003).

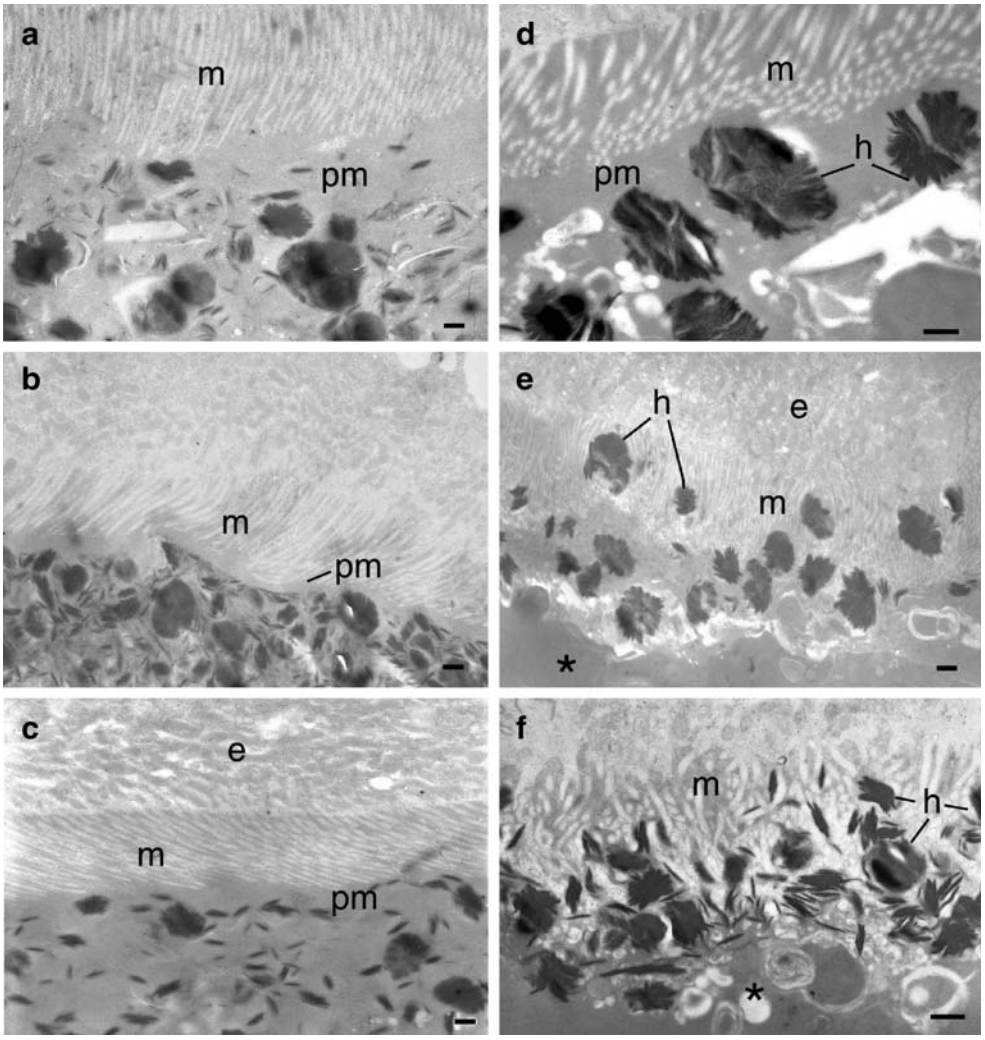

Fig. 3 TEM micrographs of midgut sections from *A. stephensi* female mosquitoes at 16 h (**a** and **d**), 20 h (**b** and **e**) and 24 h (**c** and **f**) after bloodfeeding. Scale bars=500 nm. **a**–**c** Mosquitoes fed with blood of Gaucher disease patient containing 5.2 mU/ml of CHIT. PM is completely filled by digestion products since 16 h from bloodfeeding. A thin fibrillar, internal portion of PM divides the blood products from microvilli (*m*) during the first two digestion phases. At 24 h from feeding, heme precipitates (*h*) directly contact microvilli (*m*) of epithelial cells (*e*). **d**–**f** Mosquitoes fed with blood containing 2,000 mU/ml of commercial chitinase from *P. falciparum*. A thin fibrillar, external portion of PM is interposed between the blood meal (*asterisk*) only in a few areas at 16 h after feeding (**d**). The products of blood digestion including heme precipitates (*h*) are clearly visible among the microvilli (*m*) and inside the epithelial cells (*e*) since 20 h

Billingsley and Rudin (1992) observed that PM-thickening partially inhibits development of *P. falciparum*, thus representing a physical barrier that parasites have to overcome by secreting a specific chitinase.

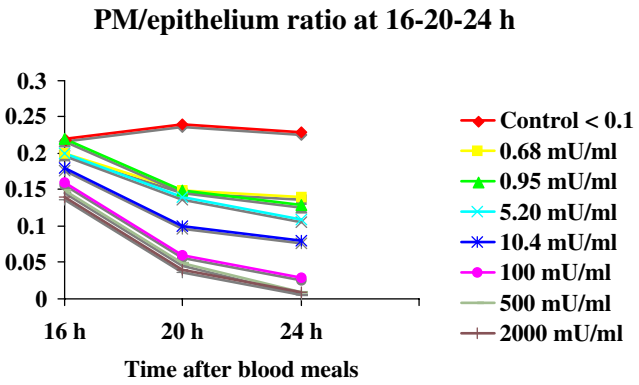

Fig. 4 Mean and range of PM thickness as measured on the microscopic section of stomach (PM/epithelium) from ten dissected midguts in presence of different concentration of CHIT

Table 1 Plasma chitotriosidase activity in several conditions (mU/ml)

Subjects	Chitotriosidase (mU/ml)	Reference
Healthy Sicilian donors	0.07 (0.016–0.42)	Malaguarnera et al. 2003
Healthy Sardinian donors	0.15 (0.02–0.41)	Malaguarnera et al. 2003
Healthy African donors	0.26 (0.016–2.11)	Malaguarnera et al. 2003
Sicilian β-thalassemia trait	0.50 (0.1–1.8)	Barone et al. 1999
Multiple sclerosis patients	0.46 (0.013–0.94)	Sotgiu et al. 2005
Acute *P. falciparum* malaria	0.77 (0.04–2.17)	Barone et al. 2003
Sicilian β-thalassemia major	0.97 (0.67–3.31)	Barone et al. 2001
Sardinian β-thalassemia major	1.20 (0.67–3.73)	Barone et al. 2001
Gaucher disease patients	35.23 (0.04–110.8)	Giraldo et al. 2001

It is known that incorporation of exogenous chitinases into a blood meal could prevent PM formation (Shahabuddin and Kaslow 1993; Filho et al. 2002). Moreover, the inhibition of the endogenous mosquitoes' midgut chitinase with allosamidin, an inhibitor of chitinase, delays PM disassembling at the beginning of blood digestion (Shen and Jacobs-Lorena 1997).

In 1993, Shahabuddin and Kaslow (1993) suggested the use of allosamidin, or specific antibodies, to reduce the malaria infection rate (Shahabuddin 1995), thus hampering the development of malaria parasites into mosquitoes' vector. This transmission-blocking way could be recognized as an available mechanism for disease control (Kaslow 1993). Nevertheless, the possibility to block the transmissibility with chitinase inhibition, with either allosamidin or specific antibodies against CHIT, is far from practical application. In fact, human chitotriosidase and plasmodial chitinase are structurally/functionally very similar because they belong to the same 18 glycosyl hydrolases family. For this reason, selective PfCHT1 inhibition (Langer and Vinetz 2001) is difficult to obtain.

Our findings support the hypothesis that human plasma CHIT may influence the *P. falciparum*/human host relationship, thus helping the parasite to cross the PM and reach the midgut epithelium. Different levels of human CHIT have been reported in healthy populations living in areas where malaria was/is endemic. Moreover, it has been observed that 5–6% of healthy Caucasian individuals are deficient in CHIT. This is due to homozygous duplication of 24 bp in exon 10 of the Chit gene (Boot et al. 1998). As a consequence, 35–45% of Caucasian individuals are heterozygotes. Therefore, a significant difference in the plasma CHIT levels in healthy individuals from Africa, Sardinia and Sicily, proportioned to the allele dosage in these populations (Malaguarnera et al. 2003), could be observed (see Table 1).

Because parasite penetration in the anopheline midgut depends on PM disruption, we may hypothesize that elevated CHIT activity levels in the plasma of African malaria patients could help *P. falciparum* to complete its life cycle in the vector, thus contributing to the malaria maintenance in endemic areas.

On the contrary, in European regions where CHIT became redundant for the expansion of inactive Chit gene, the rapid eradication of malaria together with the use of DDT and of land reclamation was possible.

It is well known that a genetic protection from malaria is conferred by heterozygotes for haemoglobin S (HbS), thus reducing the circulating asexual parasitemia (Luzzatto et al. 1970). On the other hand, HbS carriers show a compensatory effect on the transmissibility of parasite from humans to mosquitoes. Robert et al. (1996) observed that the percentage of *Anopheles* infected by HbAS gametocyte carriers was higher compared to mosquitoes fed with HbAA carrier blood. This increase in parasite transmissibility could depend on higher CHIT levels in the HbAS carrier blood due to an increased red cell turnover and a CHIT macrophage overproduction in African population (Malaguarnera et al. 2003). In Sardinia and Sicily, a similar phenomenon could occur, induced by the different prevalence of β-thalassemia or G6PDH deficiency. An adaptive strategy of the malaria parasite, based on a selective advantage in Africa (Vinetz 2005), could be hypothesized. This could have developed a significant epidemiological effect by increasing the global level of malaria transmission in countries as Burkina Faso through a genetic characteristic of population that maintains a high CHIT production in activated macrophages (Malaguarnera et al. 2003).

In conclusion, our results support the hypothesis that, in malaria patients, high CHIT levels could help *P. falciparum* to complete its life cycle in *Anopheles* midgut. We believe that this hypothesis should be further confirmed by experimental infections with blood samples from *P. falciparum* malaria patients at different chitinase concentrations.

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